Fractions and the number line

1

a $\frac{3}{4}$ b $-\frac{8}{5}$ c $7\frac{1}{2}$ d -1

e $-1\frac{1}{10}$ f $2\frac{1}{4}$

2

a $-\frac{1}{7}$ b $\frac{7}{13}$ c $\frac{2}{11}$ d $-\frac{1}{4}$

3

a $\frac{12}{5}$ b $2\frac{1}{2}$

4

a $-\frac{16}{3}$ b $-3\frac{1}{4}$

Fractions with different denominators

5

a $-\frac{5}{3}$ b $\frac{2}{3}$ c $-\frac{9}{16}$ d $\frac{7}{2}$

6

a $-\frac{7}{5}$ b $-\frac{7}{5}$ c $\frac{7}{2}$ d $-\frac{15}{8}$

7 $-\frac{8}{16}, -\frac{3}{16}, -\frac{2}{16}$

8 $-\frac{3}{14}, -\frac{6}{14}, -\frac{19}{14}$

9

a i $-\frac{8}{24}, -\frac{15}{24}, -\frac{7}{24}$ ii $-\frac{5}{8}, -\frac{1}{3}, -\frac{7}{24}$

b i $-\frac{20}{40}, -\frac{24}{40}, -\frac{7}{40}$ ii $-\frac{7}{40}, -\frac{2}{4}, -\frac{6}{10}$

c i $-\frac{24}{14}, -\frac{39}{14}, -\frac{38}{14}$ ii $-\frac{39}{14}, -\frac{19}{7}, -1\frac{5}{7}$

d i $-\frac{18}{45}, -\frac{30}{45}, -\frac{11}{45}$ ii $-\frac{6}{9}, -\frac{2}{5}, -1\frac{11}{45}$

Arithmetic with directed fractions

10

a $\frac{22}{5}$ b $\frac{11}{5}$ c $-\frac{1}{4}$ d $-\frac{1}{4}$

e $\frac{11}{16}$

11

a $\frac{10}{3}$ b $-\frac{10}{3}$ c $-\frac{11}{5}$ d $-\frac{4}{11}$

e $-\frac{16}{9}$ f $2\frac{7}{9}$ g $\frac{17}{45}$ h $-\frac{7}{9}$

Applications

12 $2\frac{1}{5}$ km to the east.

13 $\frac{7}{16}$

14 $-\frac{4}{15}$ m

15 $1\frac{13}{24}$ kg

16

a $21\frac{1}{4}$ cups

b Yes, because they sold 17 cups which earned them $34 which is larger than $18.

17 $5\frac{1}{2}$ kg